

Life Insurance Distribution & Sourcing Dashboard

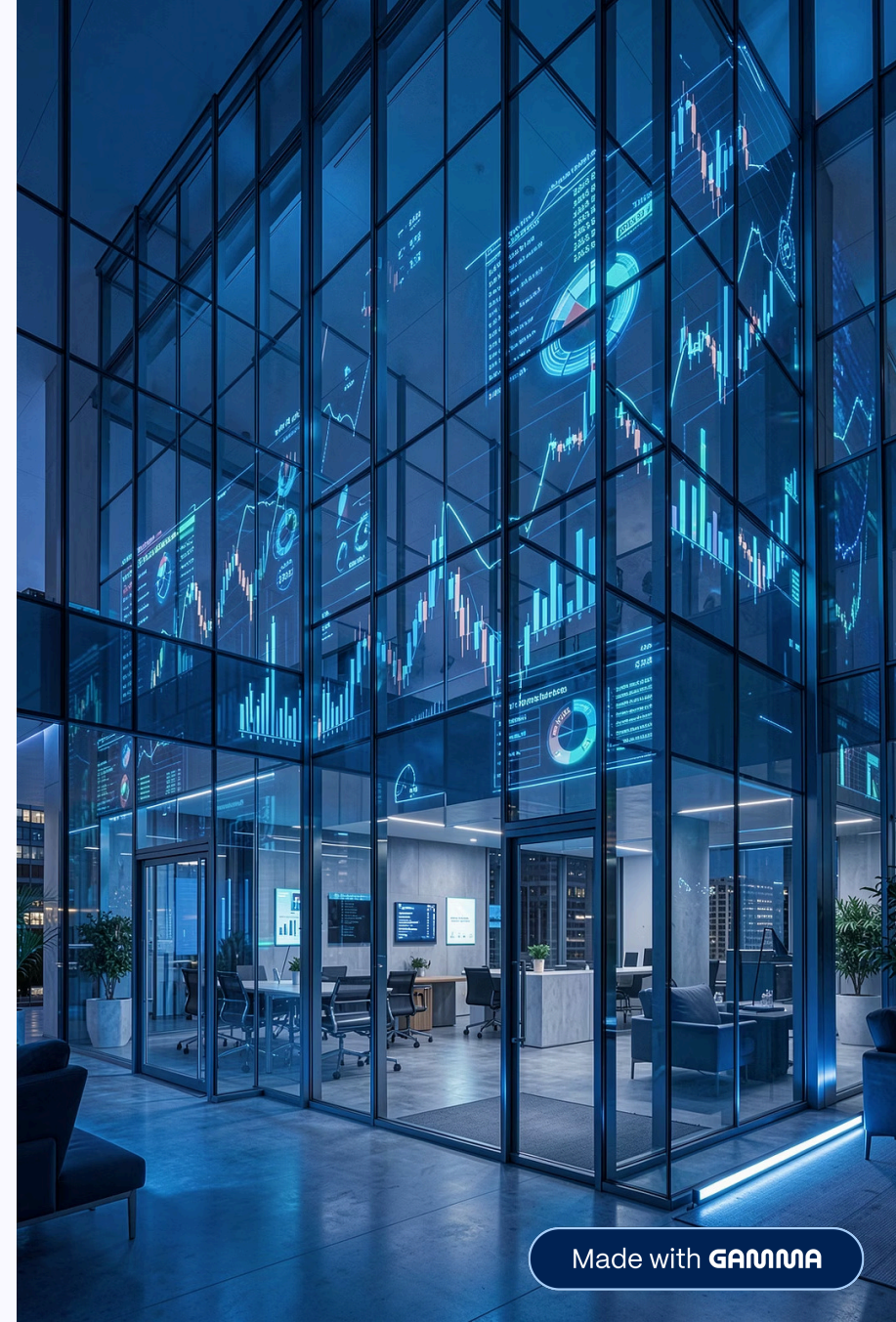
An end-to-end data engineering and business intelligence pipeline evaluating the performance, costs, and quality of a life insurance agency distribution network at scale – transforming raw operational data into actionable strategic insights.

DATA ENGINEERING

POWER BI

SQL SERVER

PYTHON



Made with GAMMA



The Business Challenge

In the life insurance industry, the agency network is the primary engine of revenue distribution. However, managing a massive direct sales force presents complex operational hurdles. Sourcing profiles must constantly balance **volume** (recruiting a high number of agents) against **quality** (ensuring those agents are productive and selling sustainable business).

Legacy data tracking systems often leave management blind to crucial questions:

Which channels yield the most profitable agents?

Which recruitment channels yield the most profitable agents?

Are high-cost channels delivering ROI?

Are high-cost channels genuinely delivering a solid return on investment?

Which channels bring in high-risk business?

And which channels are inadvertently bringing in high-risk business that results in early policy lapses?

Project Objective

What We Built

This project establishes an end-to-end data engineering and business intelligence pipeline to evaluate the performance, costs, and quality of an agency distribution network at scale.

By analyzing a comprehensive data model consisting of **20,000 unique agents** and **500,000 historical policies**, this dashboard transforms raw operational data into actionable strategic insights. It empowers distribution heads to optimize recruitment budgets, track agent activation velocity, and safeguard the company's bottom line against revenue leakage from lapsed policies.

20K

Unique Agents

Distinct producing agents in the dataset

500K

Historical Policies

Individual policies tracked across the network

300.51

Company-Wide ROI

Baseline ROI multiplier across all channels

Core Pillars of the Pipeline

To demonstrate production-level capabilities, the project was executed across four distinct phases:

1

Data Engineering & Simulation

Synthesized a multi-table relational schema using Python (pandas, Faker) to replicate real-world insurance structures, embedding realistic industry skews and intentional data corruption (duplicates, nulls, timeline anomalies).

2

ETL & Data Cleaning

Built a strict programmatic cleaning pipeline in Python to resolve structural defects, normalize text fields, strip complex string constraints from financial columns using Regular Expressions, and enforce chronological date integrity.

3

Relational Database Modeling

Bulk-loaded the structured assets into Microsoft SQL Server, establishing strict referential integrity (FOREIGN KEY constraints) and executing performance queries to benchmark operational metrics.

4

Executive Dashboard Design

Connected the relational warehouse to Power BI, designing a clean visual hierarchy. Developed custom analytical calculations utilizing advanced DAX measures to track dynamic, cross-filtered recruitment ROI and portfolio health.

Data Architecture & Summary

This project is built on a synthetic, relational dataset designed to mirror the actual operational scale and metrics of a life insurance distribution network. The dataset consists of two primary tables linked via a one-to-many relationship (Agent_ID).

Agency Distribution Data (Agents Table)

Total Records: 20,000 unique agents. Represents the complete sales force and the recruitment metrics associated with onboarding each individual.

- **Sourcing_Channel:** Agency Partner, LinkedIn, Referral, Job Portal, Walk-in
- **Onboarding_Date**
- **Recruitment_Cost:** The capital spent to acquire the agent

Book of Business Data (Policies Table)

Total Records: 500,000 individual policies. Represents the cumulative premium generation and retention risk driven by the agency network.

- **Policy_Type:** Term Life, ULIP, Whole Life, Endowment
- **Status:** Active, Lapsed, Surrendered, Unknown
- **Issue_Date**
- **Premium_Amount:** The top-line revenue generated per policy

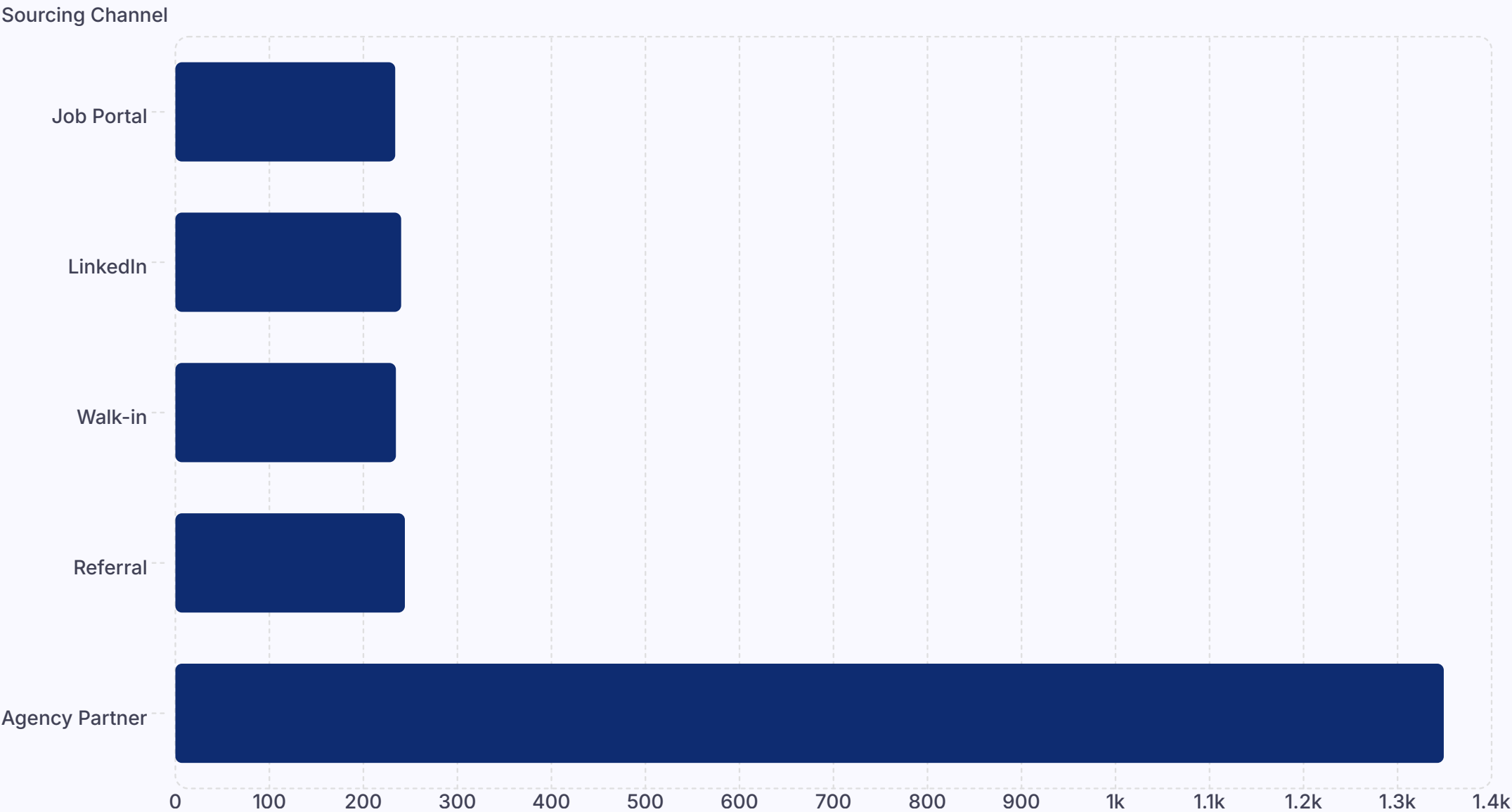
Overall Channel Volume & Value

Which sourcing channel recruits the most agents, and which brings in the most total premium? The table below shows Total Agents Recruited, Total Policies Sold, and Total Premium Generated across all five sourcing channels.

Sourcing Channel	Agents Recruited	Policies Sold	Premium Generated
Agency Partner	4,062	101,220	₹5,591,011,000
Referral	4,020	100,860	₹5,583,846,000
LinkedIn	3,987	99,937	₹5,543,457,000
Job Portal	4,013	100,103	₹5,541,452,000
Walk-in	3,918	97,880	₹5,439,309,000

Sourcing ROI: Return on Investment

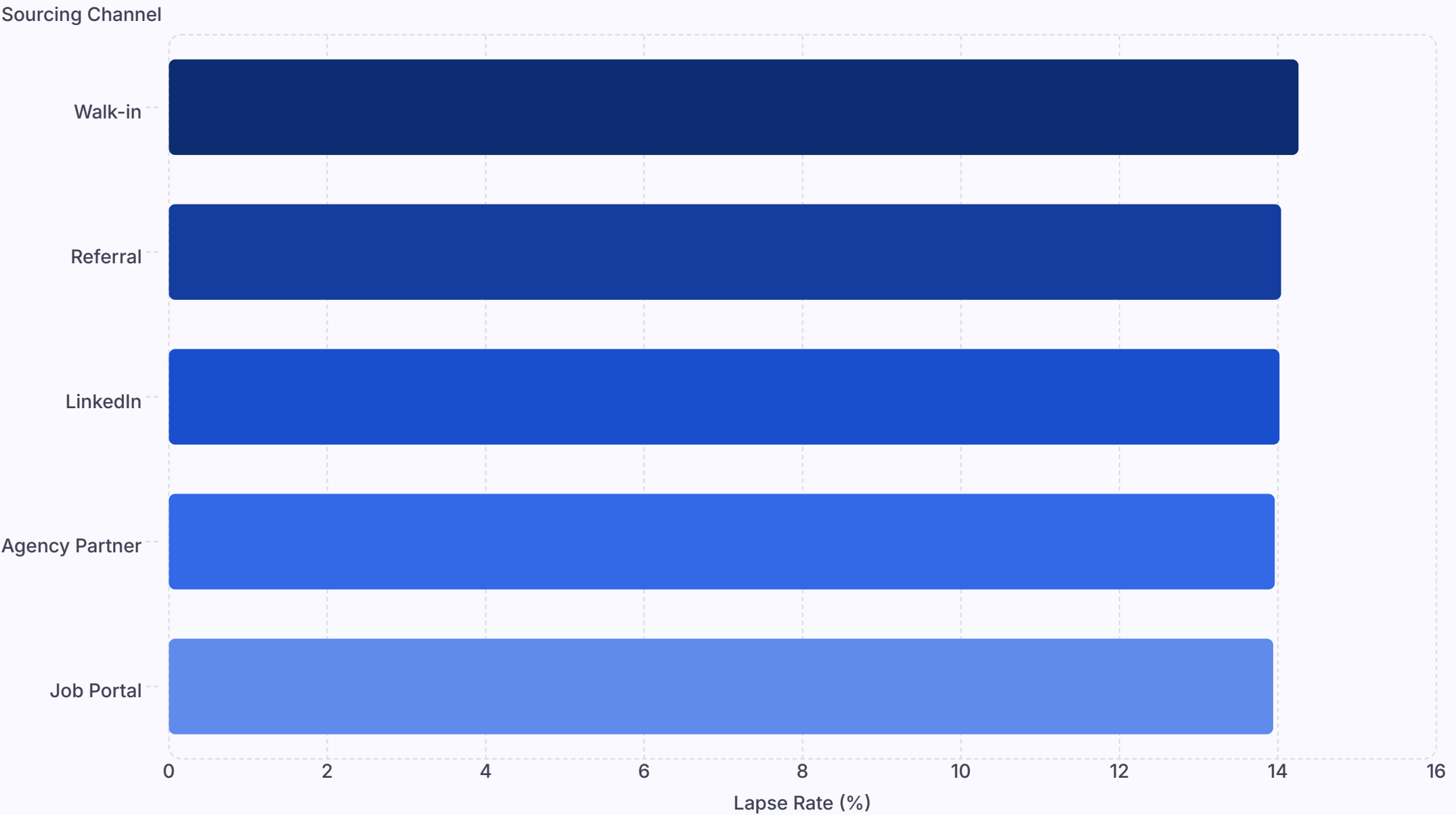
The Business Question: We spend different amounts recruiting on LinkedIn vs. Job Portals. Which channel gives us the highest premium return per rupee spent on recruitment? This analysis proves the value of calculated columns and understanding business profitability. The custom DAX measure divides total premium generated by total recruitment cost per channel.



❏ Agency Partners drive the highest gross premium (₹5.59bn) but at a dramatically higher recruitment cost (₹1,349M), resulting in the lowest ROI multiplier of 4x – compared to 22- 23x for all other channels.

Agent 30-Day Activation & Policy Lapse Rates

The Business Question: In agency management, if an agent doesn't sell quickly, they usually quit. What percentage of our agents sold their first policy within 30 days of onboarding? This analysis uses advanced date manipulation (DATEDIFF), Common Table Expressions (CTEs), and complex logic. The dashboard also highlights that ~11.27% (₹3.12bn) of the portfolio is currently categorized as "Lapsed."



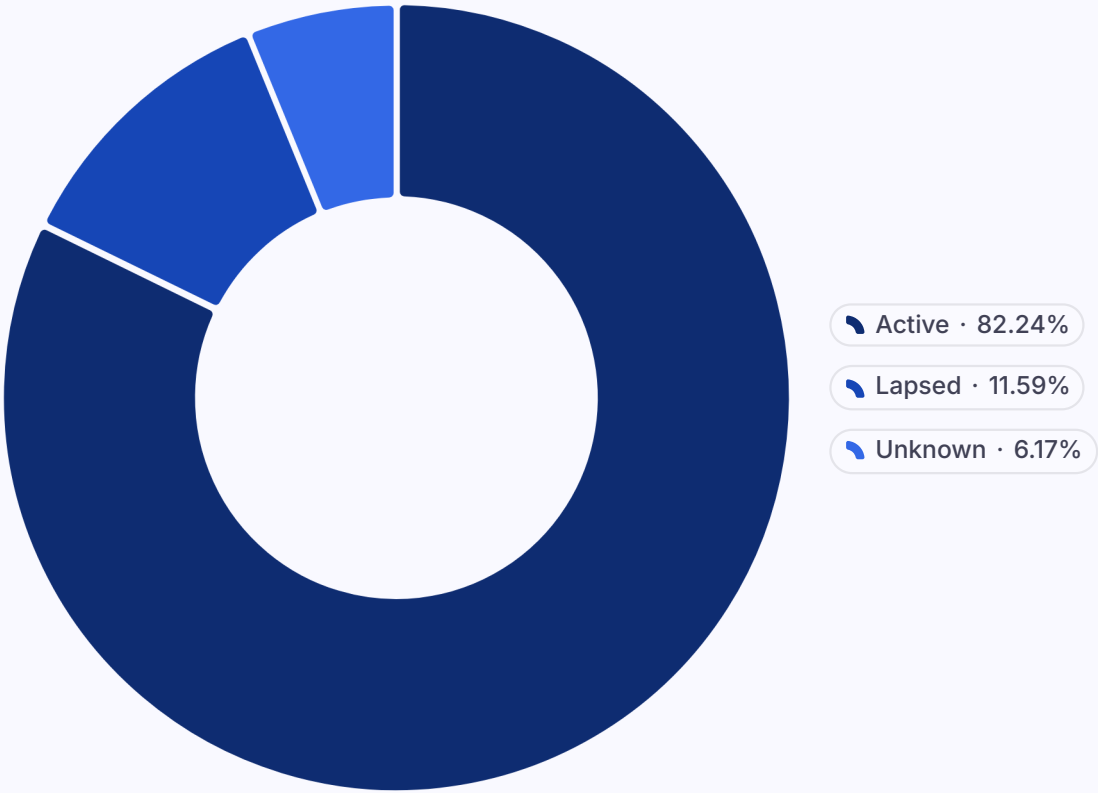
Walk-in channels show the highest lapse rate at 14.26%, while Job Portals have the lowest at 13.94%. The business must utilize the cross-filtering capability of this dashboard to identify which specific Sourcing Channels are driving these lapses. Channels with high lapse rates should face stricter quality-of-business checks or commission clawback structures to prevent revenue leakage.

Power BI Dashboard: Portfolio Health

Portfolio KPIs

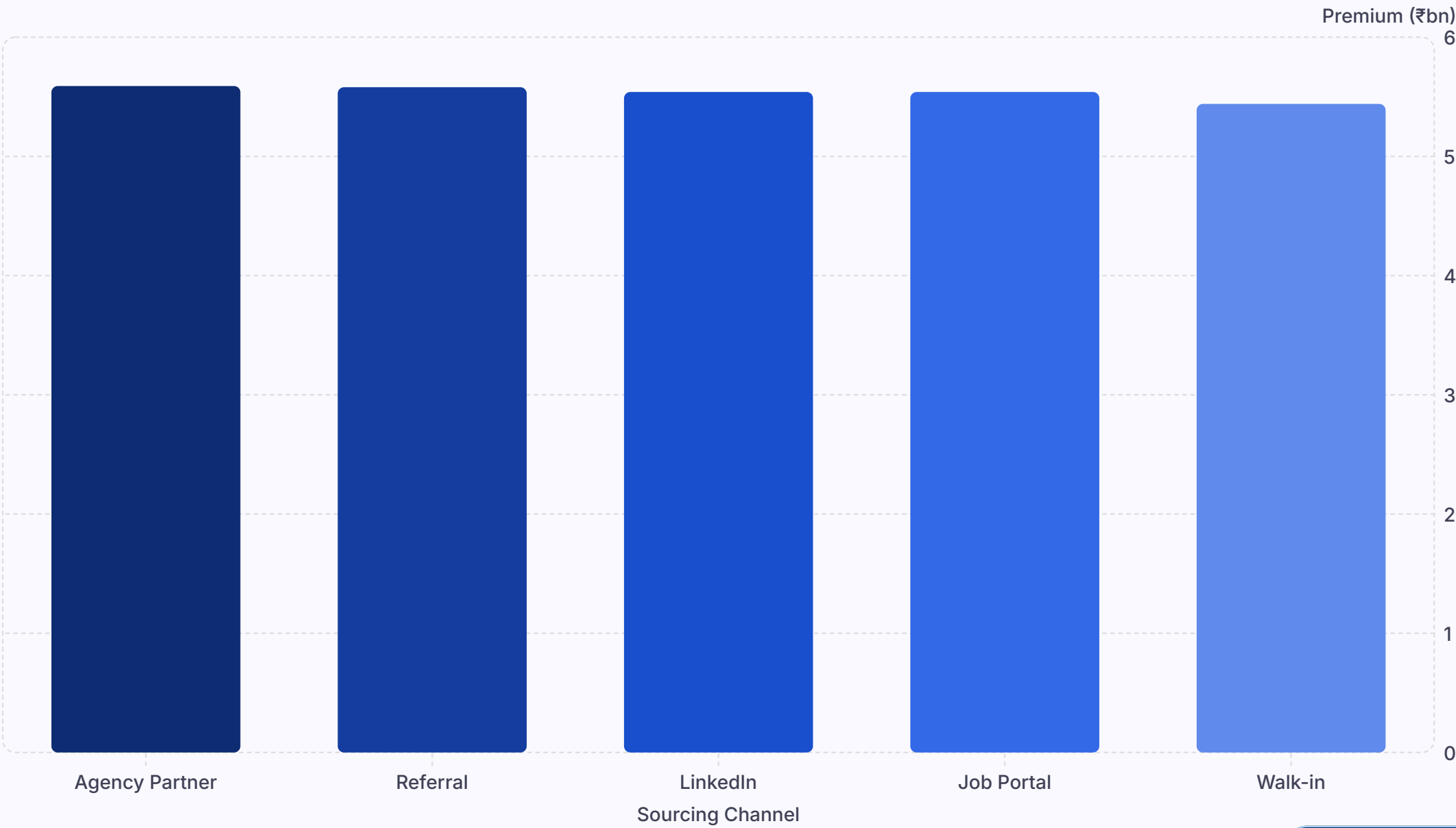


Premium by Policy Status



Active policies represent 79.94% (₹22.14bn) of the total portfolio. **Lapsed policies** account for 11.27% (₹3.12bn), and **Unknown status** for 5.99% (₹1.66bn). The business must utilize the cross-filtering capability of this dashboard to identify which specific Sourcing Channels are driving these lapses.

Total Premium by Sourcing Channel



Strategic Business Recommendations

Based on the end-to-end data analysis, the following strategic actions are recommended to the Distribution & Agency Management teams:

1 Reallocate Recruitment Budget Based on ROI, Not Volume

While Agency Partners drive the highest gross premium volume, the custom DAX calculation reveals variations in the ROI Multiplier. The business should shift a larger percentage of the recruitment budget toward high-ROI, low-acquisition-cost channels (such as Referrals or Walk-ins) to maximize profitability per agent.

2 Implement Quality Control for High-Lapse Channels

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3 Accelerate 30-Day Activation via Targeted Training

The SQL analysis of "30-Day Activation Velocity" identifies which channels bring in "fast starters." Training and onboarding resources should be aggressively targeted toward agents from slower-activating channels to get them producing within their first month, reducing agent attrition.

4 Align Product Strategy with Sourcing Channel

By utilizing the Policy_Type slicer (Term, ULIP, Endowment, Whole Life), management can identify product-specific trends. If agents from LinkedIn are proven to sell more high-margin ULIPs, recruitment messaging on that platform should be specifically tailored to attract candidates with financial advisory or investment backgrounds.